



**NANYANG
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SINGAPORE

**FROM STRUCTURED SUPERVISION TO OPEN-ENDED
REASONING: AN EVALUATION FRAMEWORK FOR
GENERALISATION AND CALIBRATION IN
PSYCHOLOGICAL QA**

Tian Yidong

College of Computing and Data Science

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Tian Yidong

Supervisor: Prof. Miao Chun Yan

Examiner: Prof. Cong Gao

College of Computing and Data Science

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Abstract

Large Language Models (LLMs) have demonstrated strong capabilities in structured natural language tasks such as classification and multiple-choice question answering. However, there remains a lack of systematic evaluation frameworks for assessing the generalisation capability of LLMs across task formats in psychologically grounded reasoning domains, which include social inference, affective inference, and human-centred open-ended question answering. Existing studies tend to investigate prompting strategies, evaluation metrics, calibration, or reasoning elicitation techniques such as chain-of-thought in isolation. However, these research did not jointly considering how prompt and answer format constraints, answer extraction robustness, and the inclusion of explicit reasoning during generation collectively influence observed model performance. This fragmentation makes it difficult to determine whether performance gains reflect genuine reasoning ability or arise from artifacts of task design and evaluation methodology.

In this work, we propose a unified evaluation framework to systematically investigate the generalisation behaviour of a QLoRA fine-tuned Qwen2.5-1.5B-Instruct model across task formats in psychological reasoning. The framework encompasses six interrelated experimental components: (1) transfer from multiple-choice training to open-ended question answering, (2) prompt constraint ablation, (3) answer extraction strategy ablation, (4) reasoning-based prompting, (5) calibration analysis via semantic self-consistency, and (6) structured error analysis. In addition to traditional lexical measures, we also incorporate semantic similarity metrics in order to address the inadequacy of exact-match evaluation in open-ended settings.

The experiments show that strong performance on MC tasks does not transfer effectively to Open QA; this indicates a substantial task-format mismatch. Prompt constraints improve output regularity and lexical scores; however they do not resolve reasoning failures. More robust extraction strategies can lead to reduction in evaluation brittleness; this shows that some apparent errors are caused by pipeline sensitivity rather than model incompetence. Reasoning-based prompting improves semantic alignment; however it does not contribute to strict lexical correctness. This suggests a trade-off between semantic adequacy and structural reliability. Furthermore, calibration analysis reveals that semantic self-consistency is positively correlated with correctness; however, it still remains imperfect as a confidence signal due to overconfident incorrect predictions. Overall, the results suggest that the main limitation of the model lies in semantic reasoning and calibration rather than output formatting alone. These findings inform that future evaluation of LLMs in psychologically grounded open-ended tasks should rely on multi-metric, reliability-aware, and extraction-robust frameworks rather than only structured-task accuracy. Additionally, the present framework is limited by its focus on a single base model, a relatively small Open QA evaluation set, and dependence on automatic semantic similarity measures, which should be addressed in future work.

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0.1 Introduction

0.1.1 Background

Large Language Models (LLMs) have demonstrated strong performance across various natural language processing tasks, including text generation, question answering, and reasoning [1, 2]. These models are typically trained on large-scale corpora. And they are further adapted through instruction tuning or supervised fine-tuning. These enable them to perform complex tasks in zero-shot or few-shot setting.

Recent studies show that LLMs demonstrate emerging reasoning capabilities; this is often shown when they are given prompted with structured formats such as chain-of-thought reasoning [3]. However, the evidence for these reasoning abilities often shown in tasks, which have constrained output spaces. These tasks include multiple-choice question answering or benchmark-style evaluations. One of the most important reasons of the LLM’s capability in these tasks may be that the output space is limited and evaluation is straightforward; it often relies on exact-match correctness.

Contrastly, open-ended reasoning tasks faces a fundamentally different challenge. These tasks require models to generate free-form answers that are semantically valid but are lexically different. In domains such as psychological reasoning, social inference, and commonsense understanding, there may be multiple acceptable answers, and correctness is often context-dependent [4, 5]. As a result, evaluation becomes more ambiguous and cannot rely solely on lexical matching.

Furthermore, the ability of LLMs to generalise across task formats remains poorly understood. Models trained under structured supervision may learn patterns that are specific to the training format; they may not develop transferable reasoning capabilities. Therefore, an important question should be proposed: whether performance gains observed under structured evaluation settings can truly reflect general reasoning ability, or they only show format-specific pattern recognition.

Understanding this distinction is critical, especially for applications involving human-centered reasoning tasks, where reliability, interpretability, and robustness are essential.

0.1.2 Motivation

Although LLMs have strong benchmark performance, more concerns have been raised from about the robustness and reliability of LLM reasoning in recent studies. Although models perform well on structured tasks, they often demonstrate degraded performance when evaluated in more flexible or open-ended settings [6, 7]. This raises concern that existing evaluation protocols may overestimate model capabilities.

One potential cause of this discrepancy are task format, prompting strategy and evaluation methodology. For example, constrained prompts can enable LLMs to produce structured outputs. This can improve extractability and exact-match scores. However, more flexible prompts may produce richer but less regular outputs. Additionally, evaluation approaches based on strict string matching may cause issue of failing to capture semantically correct answers have different lexical information.

Another important factor to be considered is model confidence. In open-ended tasks without strictly defined answers, it is crucial to assess the ability of models to reliably estimate their own uncertainty. Recent work on self-consistency has proposed to use agreement across multiple generations to represent confidence [8]. However, the extent to how this method measures align with correctness, is still a question to be considered.

Although prior work has explored prompting strategies, evaluation methods, and calibration techniques, these factors are often studied in isolation. Realistically, prompting approaches such as chain-of-thought and self-consistency have been shown to improve reasoning performance [3, 8]; however, these studies focus mainly on prompt design without systematically accounting for task format or evaluation effects.

Meanwhile, evaluation metrics such as Exact Match and token-level F1 [9] is still widely used despite their sensitivity to lexical variation. However, embedding-based metrics [10] offer improved semantic coverage, but they are not consistently integrated into evaluation pipelines.

Furthermore, recent work has highlighted that apparent reasoning performance may not reflect true generalisation. Those works state that reasoning performance heavily depends on prompt structure and task setup [6, 7].

In addition, language models are known to exhibit poor calibration, often producing overconfident incorrect predictions [11].

Despite these advances, relatively little work has systematically examined how task format, prompting strategies, answer extraction, and calibration jointly influence observed model performance. This fragmentation makes it difficult to determine whether improvements arise from genuine reasoning ability or from artifacts of prompt design, output structure, or evaluation methodology.

This gap motivates the need for a unified experimental framework that disentangles these interacting factors. We aim to better understand the true limitations of LLMs in open-ended reasoning tasks by disentangling these factors.

0.1.3 Research Questions

This work aims to address the following research questions:

1. To what extent can a model trained on structured tasks generalise to open-ended task formats?
2. How do prompt constraints affect output structure and evaluation performance?
3. How sensitive are evaluation results to the choice of answer extraction strategy?
4. Does explicit reasoning improve semantic correctness in open-ended settings?
5. Can semantic self-consistency serve as a reliable proxy for model confidence?
6. What are the dominant categories of errors in open-ended reasoning tasks?

0.1.4 Contributions

This work makes the following contributions:

- A systematic empirical study of generalisation across task formats in LLMs.
- A unified ablation framework that disentangles formatting effects from reasoning limitations.
- An evaluation pipeline analysis highlighting the impact of extraction strategies.

- A multi-metric evaluation framework combining lexical and semantic measures.
- A calibration analysis based on semantic self-consistency.
- A structured error taxonomy for open-ended reasoning failures.

0.1.5 Paper Structure

The remainder of this paper is organised as follows. Section 2 reviews related work on reasoning, generalisation, evaluation, and calibration in LLMs. Section 3 describes the evaluation framework. Section 4 outlines the experiments. Section 5 presents the results and analysis. Section 6 covers the discussions. Section 7 concludes the paper and suggests directions for future work.

0.2 Related Work

0.2.1 Reasoning in Large Language Models

Recent work has explored the reasoning capabilities of large language models, especially through prompting strategies such as chain-of-thought prompting [12]. These approaches encourage models to generate intermediate reasoning steps, which has been shown to improve performance on arithmetic, commonsense, and symbolic reasoning tasks.

However, there has been persist debate and discussion about these such improvements can reflect genuine reasoning or pattern imitation learned from training data [13]. Models may discover and learn superficial patterns in prompts rather than developing robust reasoning mechanisms. This raises concerns about generalisation and reliability.

0.2.2 Task Format and Generalisation

Instruction tuning and supervised fine-tuning are often used to adapt LLMs to specific tasks [14]. Although these methods improve the LLMs’ performance on target tasks, they may also introduce format dependence. The reason is that the models may become specialised to the structure of training data.

Recent studies suggest that models trained on classification-style tasks may face decrease in their performance when deployed in open-ended generation settings [7]. This shows the importance of evaluating cross-format generalisation, rather than solely relying on in-format performance.

0.2.3 Open-Ended Evaluation

Evaluation of open-ended tasks is still a significant challenge. Traditional metrics such as Exact Match and token-level F1 are sensitive to lexical variation. This may cause them to underestimate performance especially when multiple valid answers exist [9].

In order to address this issue, embedding-based semantic similarity metrics have been proposed to capture meaning beyond surface form [15]. These methods allow for more

flexible evaluation, but introduce additional design choices, for example, embedding model selection and similarity thresholds.

0.2.4 Confidence, Uncertainty, and Calibration

Calibration refers to the alignment between model confidence and correctness. Poor calibration can lead to overconfident errors. These errors are problematic in high-stakes applications.

Self-consistency is a method that has been proposed to improve reasoning and estimating confidence; this is achieved by sampling multiple outputs and aggregating them [16]. Although consistency is often correlated with correctness, it is not a perfect proxy, and models may still produce consistently incorrect answers.

0.2.5 Domain-Specific Reasoning

In psychologically grounded tasks, such as emotion recognition and social inference, reasoning is ambiguous and context-dependent [17, 5]. The reason is that for these tasks, it requires the models to interpret implicit cues and human mental states. And this makes evaluation more challenging than in standard QA benchmarks.

0.2.6 Research Gap

Previous work mostly focuses on studied reasoning, prompting, evaluation, and calibration in isolation. However, limited studies systematically discover the joint effect of task format, prompt constraints, extraction robustness and calibration on observed performance under open-ended reasoning task settings. This work aims to bridge this gap through a unified experimental framework.

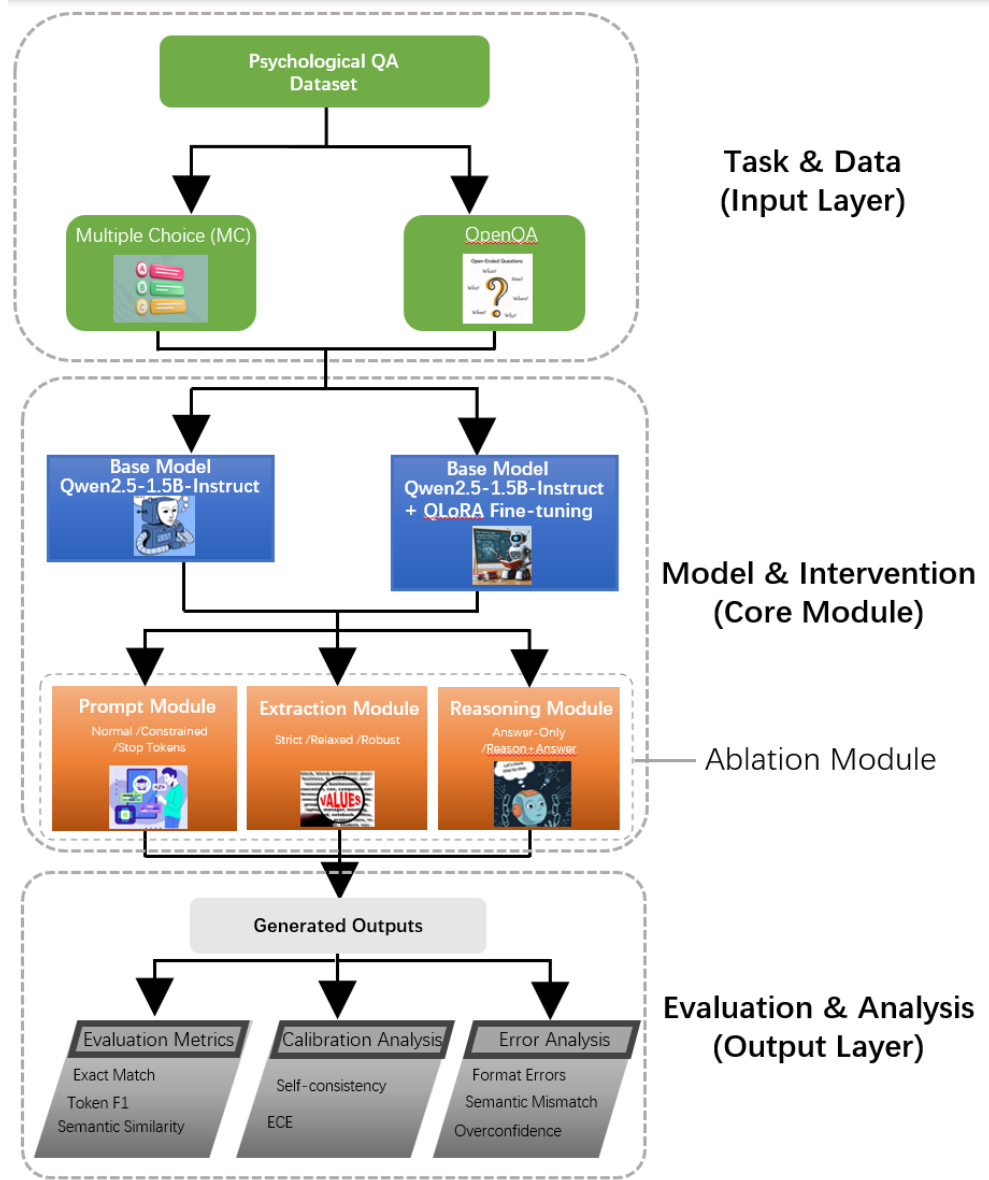


Figure 1: The framework contains three modules: Task and Data as the input layer; Model and Intervention as the core module; Evaluation and Analysis as the output layer. Task and Data module (input layer) includes: psychological-based datasets used include the Multiple-Choice (MC) and OpenQA which are psychological reasoning datasets structured in multiple choice format and Open-ended format. In the Model and Intervention module (core module), it includes base pretrained model (Qwen2.5-1.5B-Instruct) and its finetuned model using one specific dataset including the MC and OpenQA; this module also includes the Ablation sub-module, which is designed for ablation experiment of prompt instructing (the Prompt Module), LLMs output extraction (the Extraction module) and explicit reasoning output enforcement using prompt (the Reasoning module). In the Evaluation and Analysis module (output module), the generated outputs are passed to evaluation using specific metrics, to calibration analysis as well as to error analysis.

0.3 Evaluation Framework

0.3.1 Evaluation Pipeline

The framework contains three modules: Task and Data as the input layer; Model and Intervention as the core module; Evaluation and Analysis as the output layer. Task and Data module (input layer) includes: psychological-based datasets used include the Multiple-Choice (MC) and OpenQA which are psychological reasoning datasets structured in multiple choice format and Open-ended format. In the Model and Intervention module (core module), it includes base pretrained model (Qwen2.5-1.5B-Instruct) and its finetuned model using one specific dataset including the MC and OpenQA; this module also includes the Ablation sub-module, which is designed for ablation experiment of prompt instructing (the Prompt Module), LLMs output extraction (the Extraction module) and explicit reasoning output enforcement using prompt (the Reasoning module). In the Evaluation and Analysis module (output module), the generated outputs are passed to evaluation using specific metrics, to calibration analysis as well as to error analysis.

0.3.2 Task Setting and Generalisation Objective

This section provides an overview of the methodology for evaluating psychological reasoning and generalisation capabilities of LLMs, as shown in Figure 1. We use Qwen2.5-1.5B-Instruct as the base model and apply QLoRA for parameter-efficient adaptation.

Rather than training from scratch, we perform supervised fine-tuning (SFT) on the pretrained model using task-specific datasets.

1. Task Formulation

We consider two task formats within the same domain of psychological reasoning: a structured multiple-choice (MC) task and an open-ended question answering (Open QA) task.

Structured Task (Multiple Choice)

In the structured task, each instance consists of a context c , a question q , and a set of four candidate answers $\{a_1, a_2, a_3, a_4\}$. The model should select the correct option from the candidate answers.

The input is in the structure:

$$x = (c, q, a_1, a_2, a_3, a_4) \quad (1)$$

The target output is a discrete label $y \in \{A, B, C, D\}$, contained in a structured format:

$$y = \text{<answer>A</answer>} \quad (2)$$

This formulation reduces the task to a constrained generation problem that is similar to classification, as the output space is limited and explicitly defined.

Open-Ended Task

In the open-ended task, each instance consists of a context c and a question q , and the model is required to generate a answer in free form.

$$x = (c, q) \quad (3)$$

The output y is an unconstrained text that remains semantically correct but may be lexically different.

Unlike the structured task, the open-ended setting introduces:

- a significantly larger output space,
- multiple lexical expressions for the same answer,
- focus more on semantic correctness instead of exact lexical matching.

This makes language tasks evaluation more challenging. It also motivates the use of

both lexical and semantic metrics.

2. Generalisation Objective

The evaluation data is used to assess both in-distribution performance and cross-format generalisation.

Generalisation setting. The evaluation includes domain-matched and format-shifted conditions. Models trained on MC data are evaluated on Open QA tasks to assess generalisation across task formats. Meanwhile, models trained on Open QA data are evaluated on Open QA to set up an upper bound under matched conditions.

Rationale. This evaluation design enables a controlled comparison between format-aligned and format-mismatched settings. This allows us to isolate the cause of performance degradation from insufficient reasoning capability or from incompatibility between training and evaluation formats.

0.3.3 Evaluation Dimensions

To systematically investigate the generalisation behaviour of fine-tuned language models across task formats, we design six experiments. Each experiment focuses on a specific factor that may influence observed performance, which includes task format, prompting strategy, answer extraction, reasoning requirements, model calibration, and error characteristics.

The overall experimental framework is designed to disentangle three key sources of performance variation:

- **Task-format mismatch:** whether structured supervision transfers to open-ended tasks,
- **Output and evaluation artifacts:** whether performance is affected by formatting or extraction,
- **Intrinsic reasoning limitations:** whether the model fails to produce semantically correct answers.

By combining quantitative metrics with controlled ablations, the experiments aim to provide a comprehensive and interpretable analysis of model behaviour.

1. Prompting Strategies

To investigate the effect of output constraints and reasoning requirements, we design multiple prompting strategies.

Standard Prompt

The standard prompt provides task instructions without enforcing a strict output format. The model is asked to answer the question directly.

This setting reflects typical instruction-tuned models. It is used as a baseline for comparison.

Constrained Prompt

In the constrained prompt setting, the model is instructed to output answer in a fixed format:

$$\langle \text{answer} \rangle \dots \langle / \text{answer} \rangle \tag{4}$$

This constraint reduces ambiguity in output structure. Therefore, it is used to facilitate downstream evaluation.

Constrained Prompt with Decoding Control

To further enforce structural consistency, decoding-level constraints are introduced by specifying stop tokens during generation.

In this setting, when the closing tag (e.g., `</answer>`) is produced, generation is terminated. This prevents the model from generating additional explanations or additional text beyond the required format.

The purpose of this design is to disentangle reasoning failure from formatting failure; in the case when performance improves under stricter constraints, errors are likely to

be caused by output misalignment rather than reasoning inability.

Answer-Only Prompt

In this setting, the model is instructed to directly produce the final answer without any intermediate reasoning.

Reasoning + Answer Prompt

In this setting, the model is instructed to first produce a reasoning trace and then provide the final answer:

$$\langle \text{reason} \rangle \dots \langle / \text{reason} \rangle \langle \text{answer} \rangle \dots \langle / \text{answer} \rangle \quad (5)$$

This setup is used to evaluate whether explicit reasoning improves semantic correctness. It also evaluates whether explicit reasoning introduces additional structural instability in the generated output.

2. Answer Extraction Strategies

Model outputs are processed using different extraction strategies to obtain the final answer after generated by LLMs. This allows us to evaluate the robustness of the evaluation pipeline.

Strict Extraction

Strict extraction only accepts outputs that exactly match the predefined format:

$$\langle \text{answer} \rangle \dots \langle / \text{answer} \rangle \quad (6)$$

If the output does not strictly follow this pattern, the prediction is treated as invalid. This approach ensures high precision but is highly sensitive to formatting difference.

Relaxed Extraction

Relaxed extraction uses heuristic rules to recover the most likely answer from partially structured or loosely formatted outputs.

- It extracts content within `<answer>` tags even if spacing or formatting is imperfect,
- It falls back to detecting explicit answer patterns (e.g., “The answer is A”),
- It identifies standalone answer tokens (e.g., A/B/C/D) when structured tags are missing.

This approach reduces evaluation brittleness while still relying on surface-level patterns.

Robust Extraction

Robust extraction further enhances relevant content extraction by integrating noise removal and semantic-aware heuristics.

In addition to the relaxed rules, it:

- removes irrelevant prefixes, suffixes, and explanations,
- prioritises the most plausible answer span when multiple candidates appear,
- applies normalization to handle variations in casing and formatting.

It helps approximating human evaluator’s interpretation of the model output. It therefore reduces evaluation artifacts caused by strict parsing rules.

3. Calibration Analysis via Self-Consistency

For calibration analysis, we use semantic similarity as the basis for defining both inter-sample consistency and semantic correctness. We then analyse the relationship between confidence and correctness using Pearson correlation, Spearman correlation, and Expected Calibration Error (ECE).

We evaluate model confidence using a self-consistency-based framework.

For each input, we sample multiple outputs $\{y_1, y_2, \dots, y_k\}$ and compute:

Consistency: average semantic similarity among sampled outputs.

Consensus answer: the answer with highest average similarity to others.

Correctness: semantic similarity between the consensus answer and the ground truth.

We then analyse:

- Pearson correlation between consistency and correctness,
- Spearman rank correlation between consistency and correctness,
- Expected Calibration Error (ECE), which is computed by binning predictions based on consistency and comparing predicted confidence with empirical accuracy.

These metrics assess whether the model’s internal agreement signal is aligned with actual correctness.

4. Evaluation Setting for Analysis

For qualitative error analysis, we adopt a fixed evaluation setting that combines constrained prompting with robust answer extraction.

In this setting, model outputs are generated under the constrained prompt format with decoding control (i.e., stop tokens); and predictions are obtained using the robust extraction strategy.

This configuration is selected because it minimises output formatting errors and extraction artifacts, which allows the analysis to focus on intrinsic reasoning failures rather than surface-level issues.

0.3.4 Evaluation Metrics

We use two types of evaluation metrics in this work: lexical matching metrics and semantic similarity metrics.

Lexical Metrics

For all main comparison experiments, we report Exact Match (EM) and token-level F1. Exact Match measures whether the predicted answer exactly matches the reference answer after normalization. The formula of EM is shown below.

$$\text{EM} = \mathbb{I}(y_{\text{pred}} = y_{\text{gold}}) \quad (7)$$

EM metric is strict and sensitive to minor lexical variations.

In addition, token-level F1 measures the overlap between predicted and reference tokens, as shown in the following formula.

$$\text{F1} = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (8)$$

where:

- Precision = overlap / predicted tokens
- Recall = overlap / reference tokens

F1 metric is more robust to lexical variations compared to EM; however it still relies on surface-level similarity.

Semantic Similarity

For selected Open QA ablation experiments, we additionally report embedding-based semantic similarity. This metric is computed as the cosine similarity between the sentence embeddings of the predicted answer and the reference answer, as shown in the formula below.

Given embeddings e_{pred} and e_{gold} :

$$\text{Sim} = \frac{e_{\text{pred}} \cdot e_{\text{gold}}}{\|e_{\text{pred}}\| \|e_{\text{gold}}\|} \quad (9)$$

It is used to better capture semantic alignment in open-ended settings; this is important when lexically different answers are still meaningfully correct.

Calibration Metrics

For calibration analysis, we use semantic similarity as the basis for defining both inter-sample consistency and semantic correctness. We then analyse the relationship between confidence and correctness using Pearson correlation, Spearman correlation, and Expected Calibration Error (ECE).

0.4 Experiments

0.4.1 Experimental Setup

1. Model

Base model. We adopt Qwen2.5-1.5B-Instruct as the base model for all experiments.

Parameter size. The model contains approximately 1.5 billion parameters.

Model type. Qwen2.5-1.5B-Instruct is an instruction-tuned causal large language model designed for general-purpose text generation and reasoning tasks.

Why chosen. This model is selected for three main reasons. Firstly, its moderate parameter size enables efficient experimentation and controlled analysis of fine-tuning effects. Secondly, as an open-source instruction-tuned model, it allows full access to weights and adaptation through parameter-efficient methods. Thirdly, it provides a suitable balance between expressive capability and computational cost, which makes it well-suited for studying generalisation behaviour under different task formats.

2. Fine-Tuning Setup

Fine-tuning method. We employ QLoRA for parameter-efficient fine-tuning.

Training objective. The model is trained using supervised fine-tuning (SFT). The purpose is to maximise the likelihood of target outputs conditioned on input prompts.

Hardware / compute. Training is conducted on GPU with mixed-precision support in order to improve computational efficiency.

Key hyperparameters. We fine-tuned Qwen2.5-1.5B-Instruct using QLoRA for parameter-efficient adaptation. Training was conducted for 2 epochs with a learning rate of $2e-4$. An effective batch size of 8 is used. The maximum sequence length was set to 384. We used an 8-bit AdamW optimizer with a cosine learning rate schedule and a weight decay of 0.01. For LoRA, we set the rank to 8, the scaling factor to 16, and the dropout to 0.05. Training was performed using mixed precision (bfloat16 or float16 depends on hardware support) with 4-bit quantization to improve efficiency.

3. Data

Training Data

The training data consists of two distinct formats: structured multiple-choice (MC) data and open-ended question answering (Open QA) data.

Multiple-choice data. The MC dataset contains 25,262 psychological reasoning samples. Each sample includes a context, a question, four candidate answers, and one correct option. The target output is formatted as a discrete label (A, B, C, or D), wrapped in a structured tag format (e.g., `<answer>A</answer>`). This format encourages the model to learn reasoning in selection basis under constrained output space.

Open-ended data. The Open QA dataset consists of psychological reasoning tasks in free form. For this dataset, the model should generate natural language answers without predefined options. Compared to MC data, this format requires the model to produce semantically appropriate responses rather than selecting from fixed candidates. Therefore, it introduces higher variability.

Preprocessing. For both datasets, inputs are standardised into a unified prompt format. MC samples are converted into textual prompts containing context, question, and answer choices; Open QA samples are organized into pairs of context and question. Target outputs are wrapped in structured tags in order to facilitate downstream extraction and evaluation. In order to remove malformed or incomplete samples, basic text normalisation and filtering are implemented.

Evaluation Data

The evaluation data is used to assess both in-distribution performance and cross-format generalisation.

Evaluation split. For MC data, a reserved evaluation split is constructed from the original dataset. For Open QA, a separate evaluation set is created by splitting data at the scenario level. This is to avoid leakage of similar contexts between training and evaluation.

Generalisation setting. The evaluation includes domain-matched and format-shifted conditions. Models trained on MC data are evaluated on Open QA tasks to assess generalisation across task formats. Meanwhile, models trained on Open QA data are evaluated on Open QA to set up an upper bound under matched conditions.

Rationale. This evaluation design enables a controlled comparison between format-aligned and format-mismatched settings. This allows us to isolate the cause of performance degradation from insufficient reasoning capability or from incompatibility between training and evaluation formats.

Data Statistics

Dataset	Format	Domain	Train Size	Test Size
Dataset A	MC	Psychological reasoning (structured social cognition)	23999	1263
Dataset B	Open QA	Psychological reasoning (open-ended social inference)	208	39

Table 1: Summary of datasets used in training and evaluation.

Prompt design

To investigate the effect of output constraints and reasoning requirements, we design multiple prompting strategies.

The standard prompt provides task instructions without enforcing a strict output format; the model is asked to answer the question directly.

In the constrained prompt setting, the model is instructed to output answer in a fixed format:

$$\langle \text{answer} \rangle \dots \langle / \text{answer} \rangle \quad (10)$$

In the reasoning + answer setting, the model is instructed to first produce a reasoning trace and then provide the final answer:

$$\langle \text{reason} \rangle \dots \langle / \text{reason} \rangle \langle \text{answer} \rangle \dots \langle / \text{answer} \rangle \quad (11)$$

Decoding strategy

To further enforce structural consistency, decoding-level constraints are introduced by specifying stop tokens during generation.

In this setting, when the closing tag (e.g., `</answer>`) is produced, generation is terminated. This prevents the model from generating additional explanations or additional text beyond the required format.

0.4.2 Task 1: Generalisation from MC to Open QA

Goal. This experiment evaluates whether a model fine-tuned on a structured task (multiple-choice) can generalise to an open-ended question answering setting.

Setup. We consider four evaluation configurations:

- Base model evaluated on Open QA (baseline capability),
- MC-finetuned model evaluated on MC (source-task validation),
- MC-finetuned model evaluated on Open QA (cross-format generalisation),
- OpenQA-finetuned model evaluated on Open QA (format-matched upper bound).

Metrics. Exact Match and token-level F1.

Key Question. Does structured supervision enable transferable reasoning, or does performance depend heavily on task format?

Interpretation Logic.

- Strong performance on MC confirms successful fine-tuning.
- If MC-finetuned performance on Open QA is similar to the base model, reasoning transfer is minimal.
- If OpenQA-finetuned significantly outperforms MC-finetuned on Open QA, the limitation is likely due to task-format mismatch rather than model capacity.

0.4.3 Task 2: Prompt Constraint Ablation

Goal. To disentangle reasoning failure from output format misalignment.

Variants.

- Standard prompt (no format constraint),
- Constrained prompt (enforcing `<answer> . . . </answer>`),
- Constrained prompt with decoding control (stop tokens).

Metrics. Exact Match, token-level F1, and semantic similarity.

Key Question. Are low scores caused by incorrect reasoning or by failure to follow the required output format?

Interpretation Logic.

- If EM and F1 increase under constraints while semantic similarity remains stable, improvements are primarily due to formatting and extraction alignment rather than improved reasoning.
- If semantic similarity also improves, constraints may indirectly support reasoning.

Expected Insight. This experiment isolates the contribution of prompt design and decoding control to observed performance.

0.4.4 Task 3: Answer Extraction Ablation

Goal. To evaluate the robustness of the answer extraction pipeline and quantify evaluation artifacts.

Variants.

- Strict extraction,
- Relaxed extraction,
- Robust extraction.

Setup. All variants operate on the same set of model outputs to ensure a controlled comparison.

Metrics. Exact Match, token-level F1, and semantic similarity.

Key Question. To what extent do evaluation results depend on the extraction strategy rather than model behaviour?

Interpretation Logic.

- Significant improvement under relaxed or robust extraction indicates evaluation brittleness.
- Minimal change suggests that errors originate from the model rather than the extraction pipeline.

Expected Insight. This experiment reveals whether low performance is partly caused by strict parsing rules.

0.4.5 Task 4: Reasoning-based Prompting

Goal. To investigate whether explicit reasoning improves answer quality.

Variants.

- Answer-only prompt,
- Reasoning + Answer prompt.

Metrics. Exact Match, token-level F1, and semantic similarity.

Key Question. Does requiring the model to produce reasoning improve answer quality?

Interpretation Logic.

- Increased semantic similarity suggests improved reasoning quality.
- Divergence between semantic and lexical metrics highlights a trade-off between semantic adequacy and output regularity.

Expected Insight. This experiment characterises the trade-off between reasoning depth and output reliability.

0.4.6 Task 5: Calibration via Self-Consistency

Goal. To assess whether the model’s internal consistency reflects its correctness.

Procedure. For each input, we generate multiple outputs and compute:

- consistency (average similarity among outputs),
- consensus answer (central output),
- correctness (similarity between consensus and ground truth).

Metrics. Pearson correlation, Spearman correlation, and Expected Calibration Error (ECE).

Key Question. Does higher consistency correspond to higher correctness?

Interpretation Logic.

- Positive correlation indicates that consistency can serve as a confidence signal.
- Weak correlation suggests unreliable self-assessment.
- High-confidence incorrect predictions indicate overconfidence.

Expected Insight. This experiment evaluates the reliability of self-consistency as a representation for model confidence.

0.4.7 Task 6: Structured Error Analysis

Goal. To identify the dominant failure modes of the model.

Error Categories.

- Malformed outputs,
- Repeated outputs,
- Semantically correct but lexically different answers,
- Overconfident wrong cases.

Method. We analyse model outputs qualitatively and categorise errors based on predefined criteria.

Key Question. What types of errors contribute most to performance degradation?

Interpretation Logic.

- High malformed rate indicates formatting issues.
- High semantic alignment despite low lexical overlap reveals limitations of surface-level evaluation metrics.
- Overconfident wrong cases reveal calibration issues.

Expected Insight. This analysis provides qualitative evidence to support quantitative findings and helps distinguish between evaluation artifacts and genuine reasoning failures.

0.5 Results and Analysis

0.5.1 Reporting Conventions

We report Exact Match (EM) and token-level F1 in all main comparison experiments. For selected Open QA ablation experiments, we additionally use semantic similarity in order to better capture meaning-level correctness in open-ended generation.

Exact Match and F1 capture lexical alignment between predictions and ground truth. Differently, semantic similarity measures embedding-based alignment, which is more suitable for open-ended evaluation.

Unless otherwise specified, all results are averaged over the evaluation set. Higher values indicate better performance for all metrics.

0.5.2 Main Quantitative Results

Table 2 summarises the core performance across task formats.

Setting	EM	F1
Base → Open QA	0.0000	0.0313
MC-finetuned → MC	0.8361	0.8361
MC-finetuned → Open QA	0.0000	0.0297
OpenQA-finetuned → Open QA	0.0000	0.1939

Table 2: Main results across task formats. The arrow shows the direction of generalization task from the finetuned dataset to the evaluated dataset. The left side of the arrow stands for the dataset that is used to finetune the model; while the right side of the arrow stands for the dataset that is used to evaluate the performance of the model.

The results show a clear discrepancy between structured-task performance and open-ended generalisation. While the MC-finetuned model performs well on the MC task, its performance on Open QA is comparable to the base model; this indicates limited transfer ability.

1. Generalisation Across Task Formats

The MC-finetuned model achieves strong performance with F1 of 0.8361 on the MC task. This confirms that fine-tuning is successful under structured supervision. How-

ever, when evaluated on Open QA, its performance drops sharply with F1 dropping from 0.8361 to 0.0297 and Exact Match remaining near zero.

Furthermore, the MC-finetuned model performs similarly to the base model on Open QA, with F1 of 0.0297 on MC-finetuned model and 0.0313 on base model. This suggests that structured supervision does not significantly improve open-ended reasoning capability.

In contrast, the OpenQA-finetuned model shows clear improvements in F1 scores of 0.1939. This demonstrates that task-format alignment is critical for performance.

These findings strongly support the hypothesis that generalisation failure is primarily driven by task-format mismatch rather than model capacity limitations.

2. Joint Ablation of Prompt Type and Extraction Strategy

We conduct a joint ablation over three prompt types and three extraction strategies, resulting in nine experimental settings.

Prompt	Extraction	EM	F1	Semantic
Normal	Strict	0.0000	0.0000	0.0000
Normal	Relaxed	0.0000	0.0171	0.0269
Normal	Robust	0.0000	0.0801	0.3616
Constrained	Strict	0.0513	0.1101	0.3252
Constrained	Relaxed	0.0513	0.1101	0.3252
Constrained	Robust	0.0513	0.1101	0.3303
Constrained + Stop	Strict	0.0513	0.0850	0.3293
Constrained + Stop	Relaxed	0.0513	0.0850	0.3293
Constrained + Stop	Robust	0.0513	0.0850	0.3293

Table 3: Joint ablation results

Prompt constraint has a dominant effect on performance. Under the Normal prompt, Exact Match remains zero across all extraction strategies. Switching to a constrained prompt immediately raises Exact Match to 0.0513.

This indicates that output format alignment is a primary bottleneck in evaluation.

Under the Normal prompt, extraction strategy significantly affects performance. Moving from Strict to Robust extraction increases token-level F1 from 0.0000 to 0.0801

and semantic similarity from 0.0000 to 0.3616. This suggests that many model outputs contain partially correct answers that are not captured by strict extraction.

Under constrained prompts, extraction strategy has minimal impact. All extraction methods yield nearly identical results. This indicates that proper output structuring eliminates dependence on post-processing of output extraction.

Adding stop tokens does not improve performance and slightly reduces F1 from 0.1101 on constrained prompt setting to 0.0850 on constrained+stop settings. This suggests that decoding constraints may truncate useful outputs in relatively early stage, without providing additional structural benefits.

3. Reasoning vs No Reasoning

Prompt Type	EM	F1	Semantic
Answer-only	0.0000	0.0216	0.2366
Reason + Answer	0.0000	0.0523	0.3245

Table 4: Reasoning ablation results

Reasoning improves semantic similarity and token-level F1. This indicates better semantic alignment. However, Exact Match remains zero; it suggests that improvements are not reflected in strict lexical metrics.

This highlights a trade-off between semantic quality and output regularity include output format.

4. Calibration Analysis

We analyse model calibration using semantic self-consistency as a proxy for confidence. For each question, multiple outputs are sampled, and consistency is computed as the average semantic similarity among generated answers. Semantic correctness is defined as the similarity between the consensus answer and the ground truth.

The scatter plot shows a positive relationship between semantic consistency and semantic correctness. Samples with higher consistency tend to exhibit higher semantic similarity to the ground truth, which aligns with the observed Pearson (0.51) and Spearman (0.41) correlations.

However, the relationship is not perfectly linear. A number of points with relatively high consistency still correspond to low semantic correctness. This informs that the model can produce stable but incorrect answers. This confirms that while consistency is informative, it is not a reliable indicator of correctness.

The calibration curve reveals a clear deviation from the ideal diagonal line; it indicates that the model is not well calibrated. In several confidence ranges, empirical accuracy falls significantly below the predicted confidence.

This discrepancy is consistent with the relatively high Expected Calibration Error (ECE = 0.2526). It suggests that the model systematically overestimates its own correctness.

In particular, some mid-confidence regions exhibit near-zero accuracy, while even high-confidence bins fail to achieve correspondingly high correctness. This demonstrates that consistency-based confidence is not reliably aligned with actual performance.

The calibration bin analysis provides a more fine-grained view of model reliability. Across multiple bins, mean confidence consistently exceeds empirical accuracy. This indicates systematic overconfidence.

Particularly, certain mid-confidence bins, for example 0.3–0.4, show near-zero accuracy although moderate confidence levels occurs. Also, even in higher-confidence bins (0.8–0.9), accuracy remains substantially below confidence.

These results emphasize the observation which is that the model’s confidence signal reflects internal agreement rather than correctness. It also informs that consistency alone is insufficient for reliable uncertainty estimation.

Metric	Value
Mean Consistency	0.3410
Mean Semantic Correctness	0.3960
Semantic Accuracy (threshold = 0.8)	0.1026
Token-level F1	0.1110
Exact Match	0.0256
Pearson (consistency vs correctness)	0.5139
Spearman (consistency vs correctness)	0.4053
ECE	0.2526

Table 5: Calibration statistics based on semantic self-consistency

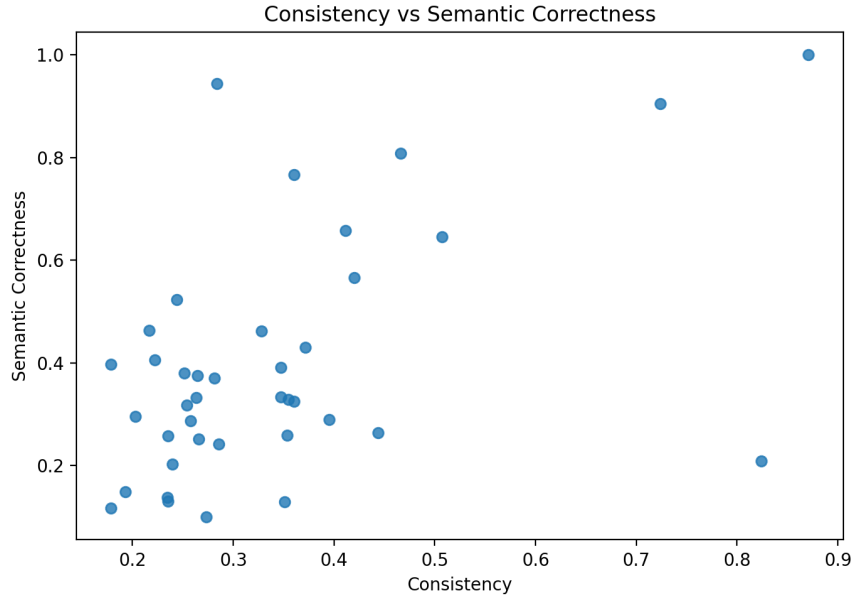


Figure 2: Scatter plot of semantic consistency versus semantic correctness. Each point represents a sample, showing the relationship between agreement among generated answers and semantic alignment with the ground truth.

Consistency and Correctness Correlation

We observe a moderate positive correlation between semantic consistency and semantic correctness (Pearson = 0.5139, Spearman = 0.4053). This indicates that samples with higher agreement among generated answers tend to be more semantically aligned with the ground truth.

This provides numerical evidence to the hypothesis that self-consistency can be used as a meaningful proxy for model confidence in open-ended QA tasks.

Partial Semantic Understanding with Low Exact Accuracy

Despite a moderate mean semantic correctness of 0.3960, the semantic accuracy under a strict threshold (0.8) is only 10.26%, and Exact Match is 2.56%. This indicates that the model frequently produces partially correct or semantically related answers, but produces fully correct responses rarely.

This gap suggests that the model possesses weak but non-trivial semantic understanding; it also suffers lacks precision and decisiveness in generating final answers.

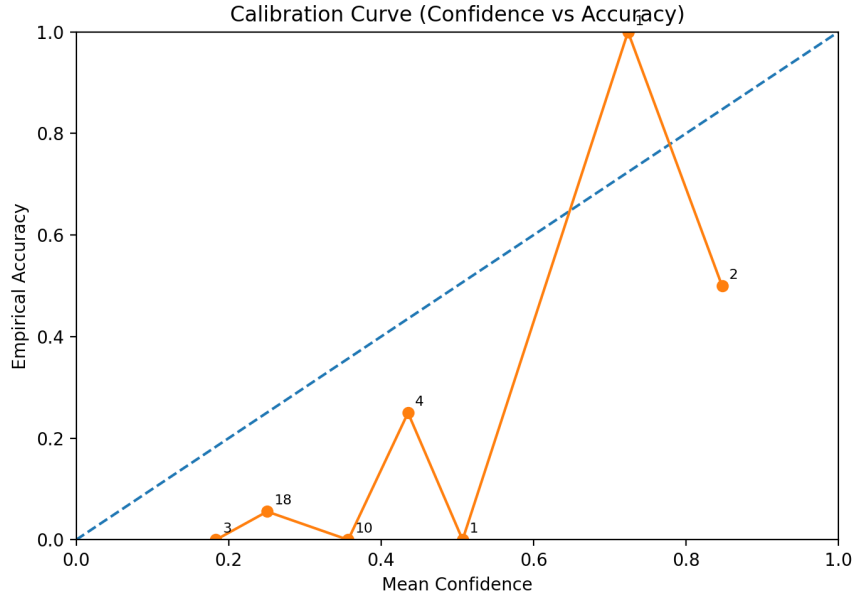


Figure 3: Calibration curve comparing mean confidence (semantic consistency) and empirical accuracy across bins. The diagonal line indicates perfect calibration.

Miscalibration and Overconfidence

The Expected Calibration Error ($ECE = 0.2526$) indicates substantial miscalibration. Analysis of calibration bins reveals systematic overconfidence. In some mid-confidence ranges (e.g., 0.30–0.40), empirical accuracy is close to zero; and even in higher confidence ranges (0.80–0.90), accuracy is far below perfect.

This shows that the model frequently assigns high confidence which is represented by consistency, to incorrect answers. This indicates that consistency alone is insufficient as a reliable confidence estimator; and this realistic analysis challenges to our hypothesis that self-consistency can be used as a meaningful proxy for model confidence.

Consistency Reflects Stability, Not Correctness

A key insight is that semantic consistency captures agreement among generated outputs rather than alignment with the ground truth.

As a result, the model can be confidently wrong (assign high consistency, low correctness), or uncertain but partially correct (low consistency, moderate correctness). This highlights a fundamental limitation of self-consistency as a standalone confidence

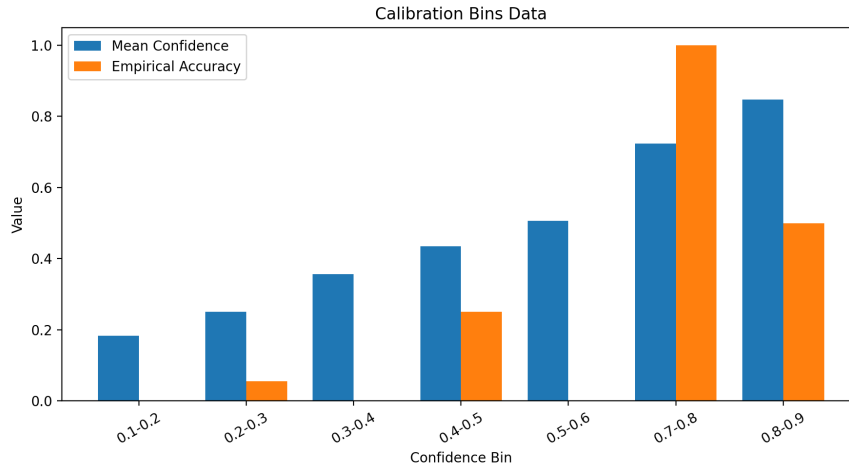


Figure 4: Calibration bin analysis showing mean confidence and empirical accuracy across different confidence intervals. Each bin aggregates samples with similar consistency levels.

signal.

Implications for Confidence Estimation

Overall, semantic self-consistency provides a useful but incomplete signal for confidence estimation in open-ended QA. Although it correlates with correctness, it fails to fully capture answer validity, especially in cases when there exists consistent but incorrect reasoning.

These findings suggest that effective confidence estimation in generative models requires combining self-consistency with additional signals, such as answer diversity or uncertainty estimation; benefit might be brought from post-hoc calibration methods.

5. Error Analysis

We categorise the observed errors into four types, as summarised in Table 6.

We categorise model errors into four main types:

- **Malformed outputs:** incorrect or incomplete formatting
- **Repeated outputs:** redundant or looping text
- **Semantically plausible but incorrect answers**

- **Overconfident wrong cases**

Table 6: Distribution of error types in the error analysis.

Error Type	Count	Ratio
Malformed outputs	0	0.0000
Repeated outputs	0	0.0000
Semantically right but lexically different	0	0.0000
Overconfident wrong cases	1	0.0256

No malformed or repeated outputs were observed. This indicates that, under the constrained prompting and robust extraction setting, output-level errors are effectively eliminated. Therefore, the remaining performance limitations can be primarily caused by deficiencies in semantic reasoning rather than formatting or decoding instability.

Absence of Semantic–Lexical Mismatch Cases

To be surprise, no cases were identified as semantically correct but lexically different. This suggests that most incorrect predictions are genuinely wrong, rather than being penalised merely because of wording differences. Therefore, the low lexical scores in this setting are unlikely to be primarily caused by evaluation mismatch.

Overconfident Wrong Cases

We identify one overconfident wrong case (2.56%). And for this case, the model produces highly consistent answers across multiple samples, whereas it remains semantically incorrect with respect to the gold reference. An example of such a case is shown in Appendix .1.

This result demonstrates that high consistency does not necessarily imply correctness, and that confidence estimation based on self-consistency can still be misleading. This finding is consistent with the calibration analysis, where consistency correlates with correctness but does not fully determine it.

0.5.3 Overall Interpretation

Across all experiments, a consistent pattern emerges. Improvements from prompting and reasoning primarily affect semantic alignment rather than exact correctness. While

constrained prompting and robust extraction significantly reduce formatting-related errors, they do not resolve the core performance limitations.

The results show that output formatting issues can be effectively controlled through prompt design and evaluation pipelines. However, even under well-structured outputs, the model continues to exhibit low Exact Match and token-level F1. This shows that the remaining errors are not due to generation instability.

Instead, the dominant source of error is incomplete semantic reasoning. The model frequently produces partially correct or semantically related answers; but it fails to reliably converge on fully correct responses in open-ended settings.

In addition, confidence estimates derived from semantic consistency are imperfect. Calibration analysis reveals that the model can produce consistent but incorrect answers, which leads to systematic overconfidence.

Overall, these findings suggest that the primary limitation of the model lies not in output formatting or decoding, but in the reliability of its semantic reasoning and confidence calibration.

0.5.4 Interim Summary

Generally speaking, the results indicate that the observed performance gap cannot be caused solely by output formatting issues or evaluation artifacts.

Although prompt constraints and improved extraction strategies lead to measurable improvements in Exact Match and token-level F1, these gains are still insufficient to bridge the gap between structured-task performance and open-ended generalisation. This suggests that although formatting alignment and evaluation robustness are important factors, they are not the primary bottlenecks.

Low lexical scores co-occur with moderate semantic similarity. This pattern indicates that the model often produces partially correct or semantically related answers that are not captured by strict lexical metrics. This shows a limitation of traditional evaluation methods in open-ended tasks.

Furthermore, calibration analysis reveals a systematic misalignment between model confidence and correctness. This can be shown by the frequent cases of overconfident incorrect predictions. Combined with error analysis, which shows minimal formatting-related failures, these findings indicate that the dominant source of error lies in semantic reasoning rather than output structure.

Overall, the results demonstrate that the primary limitation of the model lies in incomplete semantic reasoning and unreliable confidence estimation, rather than output formatting or decoding instability.

0.6 Discussion

0.6.1 Main Takeaways

This study provides several key insights into the behaviour of fine-tuned language models when transferring from structured supervision to open-ended reasoning tasks.

First of all, strong performance on structured multiple-choice (MC) tasks does not persist when generalized to effective open-ended reasoning. Although the MC-finetuned model achieves high performance on the source task, it performs similarly to the base model when evaluated on open-ended QA. This suggests that structured supervision primarily teaches task-specific output patterns rather than transferable reasoning capabilities.

Secondly, lexical evaluation metrics such as Exact Match and token-level F1 substantially understate model performance in semantically flexible tasks. The extraction ablation shows that improvements in extraction strategy significantly increase token-level and semantic scores under unconstrained prompts (normal prompts). This indicates that part of the observed failure originates from evaluation parts rather than model inability.

Furthermore, prompt engineering can resolve formatting issues effectively but it does not address underlying reasoning limitations. Constrained prompts contribute to immediate improvements in Exact Match by enforcing structured outputs, while semantic similarity remains largely unchanged. This demonstrates that formatting alignment and

reasoning quality are independent.

In addition, reasoning prompts improve semantic alignment but may reduce structural consistency. The results show that prompting the model to generate reasoning leads to higher semantic similarity and F1 score; however it does not improve the score of Exact Match proportionally. This suggests that reasoning enhances partial understanding but may introduce variability in output structure.

Finally, semantic self-consistency provides a useful but imperfect signal of model confidence. Although consistency is positively correlated with semantic correctness (Pearson = 0.5139), calibration analysis reveals systematic overconfidence. The model frequently produces consistent but incorrect answers; this indicates that agreement among generations reflects stability rather than correctness.

0.6.2 Theoretical Implications

These findings have several implications for understanding generalisation and reasoning in large language models.

Firstly, the results challenge the assumption that task performance reflects general reasoning ability. The lack of transfer from MC to open-ended QA suggests that models trained on structured supervision may rely on pattern recognition tied to specific output formats, rather than learning abstract reasoning strategies.

Furthermore, the distinction between reasoning and pattern imitation becomes evident. The improvement in semantic similarity under reasoning prompts indicates that models can generate semantically richer responses when it is encouraged to reason. However, there is an absence of corresponding gains in exact correctness. And this suggests that reasoning is often incomplete, or it may be heuristic-driven rather than fully grounded.

Additionally, the study highlights the importance of task format in shaping model behaviour. There is a strong dependence on output format. This shows that generalisation is not fully a function of model capacity; it is also an alignment between training and evaluation formats. This emphasizes the view that LLM capabilities are highly context-dependent.

0.6.3 Methodological Implications

There are important methodological implications of our work for evaluating open-ended reasoning.

Firstly, multi-metric evaluation is essential. Relying on particular limiting metrics such as Exact Match or token-level F1, can lead to misleading conclusions; this is most essential in semantically flexible domains. The inclusion of semantic similarity reveals meaningful partial correctness that may not be captured by lexical metrics.

Secondly, evaluation pipelines must be robust to output variability. The use of extraction ablation in our method, shows that different extraction strategies can largely influence reported performance; in our experiment, it is particularly obvious under unconstrained prompts. This shows that it is required to design and report evaluation pipelines carefully.

Additionally, calibration analysis should be considered while utilizing accuracy metrics. In traditional evaluation, correctness measure are used most frequently; however as demonstrated in our results, the models can be confidently wrong. Using calibration metrics such as consistency–correctness correlation and Expected Calibration Error (ECE) can provide more insights in model reliability.

0.6.4 Practical Implications

From an application point of view, experimental findings have several implications for deploying language models in real-world settings.

At first, performance on structured benchmarks should not be interpreted as evidence of reliable open-ended reasoning. Systems that are trained on classification-style tasks may have degradation when they are deployed in free-form settings without constrained output space.

Furthermore, it is important not to evaluate open-ended reasoning systems using only exact-match metrics. The reason is that these metrics may penalise valid alternative expressions and underestimate model capability. This problem is particularly serious

in domains where ambiguity or subjective interpretation exist.

Additionally, reliability and calibration are especially important judgment to be considered in high-ambiguity domains such as psychological reasoning. The presence of overconfident incorrect answers suggests that model outputs should be treated with caution. Also, additional mechanisms such as uncertainty estimation or verification are necessary for safe deployment.

0.6.5 Limitations

There exists several limitations in this study.

Firstly, the open-ended evaluation dataset is relatively small, which may limit the statistical stability of the results and the generalisability of the findings.

Secondly, the study focuses on a single base model (Qwen2.5-1.5B-Instruct). The observed behaviours may differ for larger models. Also, models with different pre-training or instruction-tuning strategies may also demonstrate different behaviours or performance.

Furthermore, the experiments are conducted within a specific domain of psychologically grounded QA; and it may not fully represent other types of reasoning tasks.

Meanwhile, semantic similarity is measured using a specific embedding model. It might be influenced by the chosen representation and may not perfectly reflect human judgment.

In addition, the study relies mainly on automatic evaluation, where there is limited human evaluation of the quality of the answer.

Finally, the analysis is constrained to a finite set of prompt designs and decoding strategies. Therefore, it may not consider and test all possible configurations.

0.6.6 Future Work

Lastly, future research can be performed from this work.

Firstly, larger and more diverse open-ended datasets should be used to improve the robustness and generalisability of conclusions.

Secondly, improved supervision strategies for reasoning, such as structured reasoning traces or step-by-step annotations, could be implemented. This may help bridge the gap between pattern imitation and true reasoning.

Thirdly, integrating reward models or verifier-based evaluation could improve answer correctness and reduce overconfident errors.

Furthermore, human-in-the-loop evaluation can provide more reliable assessment of semantic correctness and reasoning quality.

Additionally, it is suggested to use cross-model comparisons, which includes larger or differently trained models. This can help determine whether the observed limitations are model-specific or general.

Finally, more advanced calibration and uncertainty estimation techniques are required to improve the reliability of model confidence in open-ended reasoning tasks.

0.7 Conclusion and Future Work

0.7.1 Conclusion

This work presents a systematic study of the generalisation behaviour of a QLoRA fine-tuned Qwen2.5-1.5B-Instruct model across task formats. It particularly focus on the transition from structured multiple-choice supervision to open-ended question answering.

Our results demonstrate that strong performance under structured supervision does not imply robust generalisation to open-ended reasoning. Although the model achieves high accuracy when evaluated in the same format as training in finetuning, its performance degrades significantly under format shift. This indicates limited transferability of learned representations.

Through a series of controlled experiments, including prompt constraint ablation, extraction ablation, reasoning analysis, calibration analysis, and error analysis, we disentangle multiple factors that may contribute to performance degradation. We find that improvements in prompting and extraction primarily affect lexical metrics such as Exact Match and token-level F1, but do not substantially improve semantic correctness. This suggests that formatting and evaluation artifacts are not the primary bottleneck.

Further analysis reveals that low lexical scores consistently co-occur with moderate semantic similarity. This consistent pattern indicates that the model often captures partial meaning but fails to produce fully correct answers. Calibration analysis shows that semantic self-consistency is positively correlated with correctness. However, based on data distributional analysis, it is still an imperfect confidence signal, since the model frequently exhibiting overconfidence in incorrect outputs.

Overall, the findings suggest that the primary limitation of the model lies in semantic reasoning and calibration, rather than output formatting or decoding instability. These results highlight the importance of evaluating language models beyond structured benchmarks, which includes adopting semantically informed and reliability-aware evaluation frameworks for open-ended tasks.

0.7.2 Final Takeaway

Strong performance on structured tasks should not be interpreted as evidence of robust reasoning ability. Reliable evaluation of open-ended reasoning requires both semantically aware metrics and careful analysis of model confidence and failure modes.

0.7.3 Future Work

Future work can extend this study along several directions. First of all, larger and more diverse open-ended datasets may provide a more comprehensive evaluation of generalisation behaviour. Secondly, utilizing explicit supervision of reasoning traces may improve the model’s ability to produce precise and verifiable answers. Furthermore, integrating verifier-based evaluation or reward modeling may help mitigate overconfident errors. In addition, exploring stronger calibration and uncertainty estimation techniques, as well as cross-model comparisons, may further improve the reliability of open-ended reasoning systems.

.1 Example of an Overconfident Wrong Case

Table 7: Overconfident wrong case identified in the error analysis.

Field	Content
Scenario	Matt decided to participate in a hackathon. However, when it was time to start their team project, he became really busy with other projects and couldn't really contribute to the project.
Question	Does Matt still hope to contribute later?
Gold answer	Possibly, but uncertain.
Prediction (consensus answer)	No
Consistency	0.6586
Semantic similarity	0.1584

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